

Project Design Phase Solution Architecture

Date	15 February 2025
Team ID	LTVIP2025TMID36086
Project Name	Enchanted Wings: Marvels of Butterfly Species
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Example - Solution Architecture Diagram:

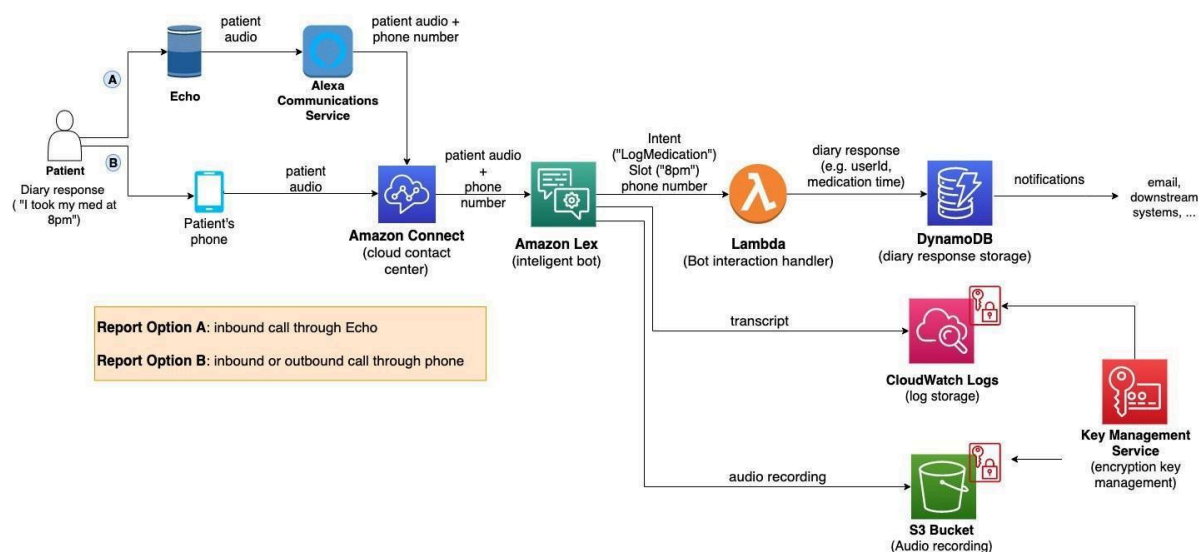


Figure 1: Architecture and data flow of the voice patient diary sample application

Reference:

<https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>

Solution Architecture bridges the gap between biodiversity-related challenges and cutting-edge deep learning solutions. It defines the overall structure and flow of how data, models, and users interact with the application.

The architecture ensures real-time butterfly classification using pre-trained CNN models, while also considering scalability, modularity, and end-user usability.

Objectives of the Architecture

- Identify butterfly species using image input from end users.
 - Use transfer learning (VGG16) for efficient, high-accuracy prediction.
 - Enable interactive and educational web-based classification interface.
 - Ensure modular design for retraining, scaling to new species, or platform deployment.
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Solution Components

1. Data Pipeline

- Input: 6499 labeled images (75 butterfly species)
- Image augmentation and preprocessing using Keras ImageDataGenerator

2. Model Architecture

- Base Model: Pre-trained VGG16 (ImageNet weights)
- Custom Layers: Flatten → Dense(256) → Dropout → Dense(75 softmax)

3. Training Environment

- TensorFlow/Keras in Jupyter Notebook/Google Colab
- GPU acceleration for training and fine-tuning

4. Model Evaluation

- Accuracy and loss plots, confusion matrix
- Final model exported (**.h5** format)

5. Deployment Layer

- Web interface using Streamlit
- User uploads butterfly image → Receives predicted species with confidence score

6. User Feedback Loop (optional future phase)

- Feedback collection for wrong predictions
- Option to submit images to improve model accuracy over time

 Architecture Diagram

